

Review

Action mechanism of rice husk ash and the effect on main performances of cement-based materials: A review

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HIGHLIGHTS

- The hydration mechanism and model of RHA were summarized and commented.
- The relationship between chemical reaction and internal curing effects of RHA was illuminated.
- The effects of RHA on main properties of cement-based materials were reviewed in detail.
- The aspects on RHA which have not been sufficiently researched were refined.

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ABSTRACT

Rice husk is a kind of agricultural waste with huge yield and wide distribution, which can get rice husk ash (RHA) containing a large amount of amorphous silica and porous structure by combusting under certain conditions. As a supplementary cementitious material (SCM), the RHA has high pozzolanic activity and internal curing characteristic. It was reported that RHA can significantly improve various performances of cement-based materials. This paper presents an overview of the physicochemical properties of RHA and its effects on the main performances of cementitious materials, e.g. cement hydration, strength, chloride penetration, sulfate resistance, carbonation and shrinkage. The hydration mechanism and the universal optimal content range of RHA are further discussed in depth. The main hydration modes of RHA particles are: (i) the dissolution reaction of micro-particles and (ii) the diffusion reaction of macro-particles. The main contribution of small-size particles to the cementitious system is the chemical effect of RHA, and the main function of large-size particles presents as the internal curing effect of RHA. Based on the summary and extraction of the macro-properties, it is considered that the preferred content range of RHA is 10–20%. Through a deeper and clearer understanding of the action mechanism of RHA in cementitious materials, it can provide ideas for further studies related to RHA and promote the green and sustainable application of RHA in cement-based materials.

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1. Introduction

The high-volume production of concrete has consumed large amounts of natural resources and increased the CO₂ emissions. The CO₂ released by one ton of cement production is about one ton, and which has a great danger to the environment [1,2]. Partially replacing cement with SCMs can reduce the consumption of cement and the CO₂ emissions related to cement industry [3]. However, conventional SCMs, e.g. FA and SF, are facing the shortage due to the limitations of production, transport, cost and resource allocation etc. RHA is a potential and economic SCM, for the high pozzolanic activity and the wide distribution of its raw material.

The raw material of RHA, rice, is one of the leading food crops, and has huge production capacity. More than 75 countries in the world cultivate rice, among them China is the largest output one. Specifically, the rice production of China was estimated at 211.2 million tons in 2016, which can produce about 42.24 million tons of rice husk, about 7.6 million tons of RHA can be obtained after combustion [4]. The rice production of China increased to 214.4 million tons in 2017 [5]. Rice husk, an inevitable by-product of rice industry, has high content of silicon and is difficult to degrade naturally. Random treatments cause potential harm to the environment, and the traditional utilization of rice husk is ineffective. In some cases, properly burning rice husk can be utilized for energy sources such as heat and electricity [6]. Use of the residual ash, i.e. RHA, as admixture in construction materials significantly increase its added value. Hu [5] prepared high active RHA by improving the combustion technique, and confirmed the environmental friendliness of RHA utilization by sustainability analysis including the embodied carbon dioxide emission and energy consumption.

At present, the influences and mechanism of RHA on the properties of cement-based materials have been widely studied. The RHA produced under a certain condition contains large amount of amorphous SiO₂ [7], which contributes to certain pozzolanic activity. The pozzolanic reaction of RHA reduces the CH content [8], thereby forming more C-S-H gels. The redistribution of water from RHA particle pores makes it have a certain role of internal curing [9]. The internal curing effect can promote the further hydration of cement, improve the strength and its development of the matrix, as well as the durability of cement-based materials to a certain extent [10,11], such as the chloride penetration resistance, sulfate corrosion, carbonation, etc. However, these studies were either not comprehensive enough or contain obvious gaps.

There is still a lack of guidance basis for the engineering application of RHA. Therefore, the purpose of this review is to integrate, refine and compare the works on the properties of cementitious materials blended with RHA. Consequently, finding out the critical problems affecting the properties of cementitious materials by RHA, to reveal the issues not covered or ignored, and to provide ideas and basis for the follow-up research and sustainable application of RHA.

The present work briefly introduces the physicochemical properties and their characterization methods of RHA. The action mechanism of RHA on cement hydration and its influence on the strength and durability of cementitious materials are mainly introduced. On this basis, the issues existing in the application of RHA in cement-based materials and prospects for the future research are put forward.

2. Properties of RHA

2.1. Physical properties

The appearance color of RHA varies from white, gray to black, which is related to its carbon content [12,13]. Nair [13] reported that the short combustion time led to insufficient combustion of RHA, thus resulting in the higher carbon content. The high carbon content is also related to K₂O. When the calcination temperature is higher than 620 °C, the potassium released by decomposition of K₂O causes the SiO₂ particles to melt and wrap the carbon particles. Thus, carbon can not contact with oxygen, resulting in black powder particles [14]. Actually, the carbon combustion is related to the contact surface of RHA with the air. If the rice husk is smashed into powder and then burned, the carbon combustion is more sufficient.

During the process of rice husk combustion, the combustion and segregation of organic matters makes the RHA particles form a porous structure [19]. As shown in Fig. 1, the RHA obtained by different scholars had different pore structures, including small size pits and large connected pores, which depend on the source of rice husk, calcination temperature, burning time, holding time, etc [20]. Madrid [21] and Habeeb [22] also observed that RHA particles, by SEM, show polygonal and angular shapes, the porous surface and honeycomb structure indicate the high specific surface area and porosity of RHA particles. Nguyen [15] attributed the higher specific surface area of RHA to its porous structure. Xu [23] also concluded that the nano-pores less than 50 nm on the

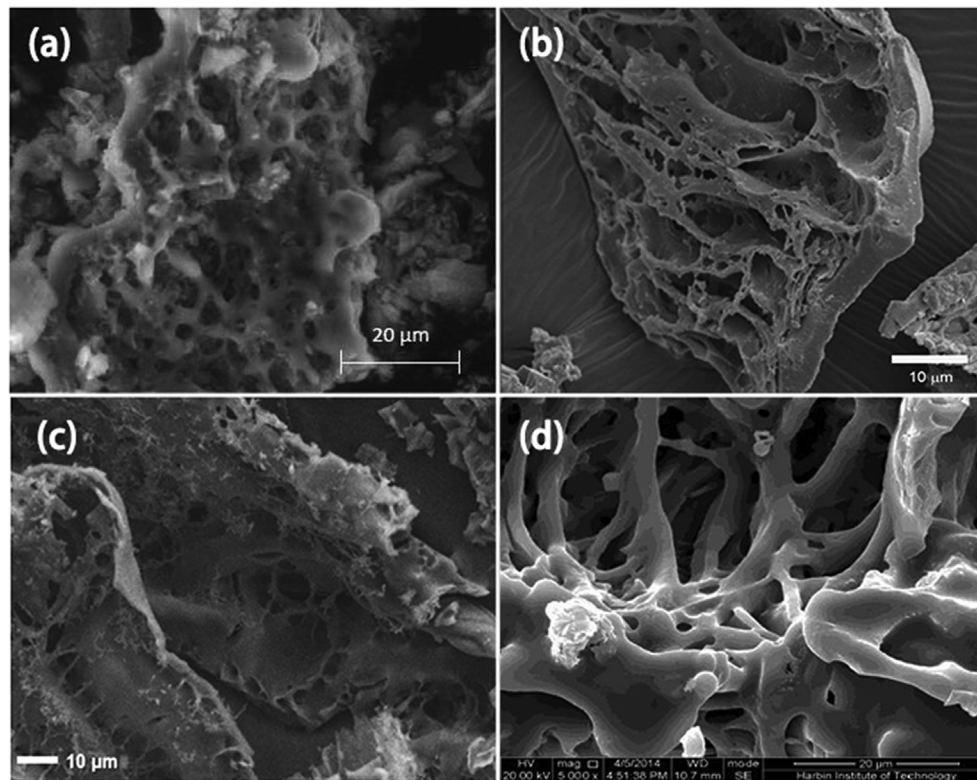


Fig. 1. SEM images of RHA particles [15–18].

Table 1
Chemical composition of RHA/%.

Source of RHA	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	SO ₃	P ₂ O ₅	K ₂ O	Na ₂ O
Finished product [27]	84.00	1.35	1.45	3.17	—	0.923	2.46	—	—
Finished product [28]	87.40	0.40	0.30	0.90	0.60	0.40	—	3.39	0.04
Finished product [29]	83.00	2.00	0.09	0.30	0.10	0.02	0.62	—	—
600 °C for 2 h [30]	86.00	1.16	1.85	0.64	0.77	0.32	0.15	0.27	1.14
650 °C for 1 h [31]	87.39	2.67	—	1.61	0.57	0.61	0.67	6.04	—
758 °C in the incinerator [32]	95.41	0.00	0.82	0.00	1.24	0.07	—	1.65	0.22
800 °C for 0.5 h [33]	93.10	0.30	0.20	1.50	0.60	—	—	2.30	0.06

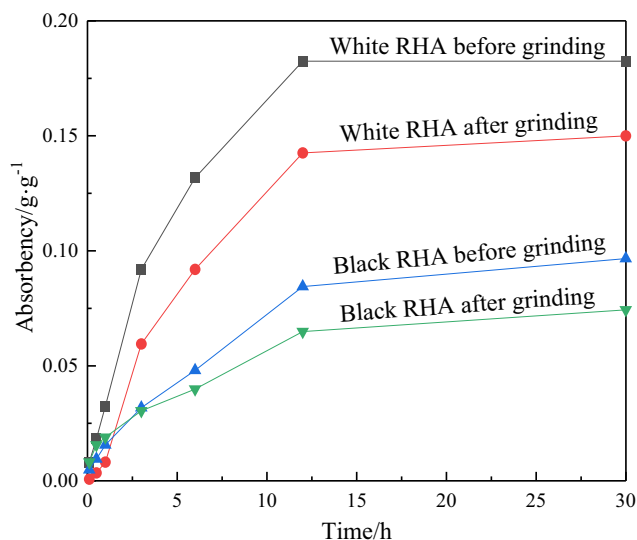


Fig. 2. Moisture absorption characteristics of RHA before and after grinding [17].

surface of RHA particles significantly increased the specific surface area.

2.2. Chemical composition and properties

The chemical properties of RHA are mainly involved in the pozzolanic reaction between SiO₂ and CH. With cement hydration going on the maximum reaction degree of RHA depends on the content of amorphous phase [15]. Rego [24] et al compared the pozzolanic activity of two kinds of RHA with high and low amorphous SiO₂ content, the results showed positive relation between RHA reaction activity and SiO₂ content. The chemical activity of RHA also depends on the morphology of SiO₂. XRD analysis shows that peak of amorphous SiO₂ appears as a wide peak at around 20°(2θ) [25]. Higher combustion temperature (800 ~ 1000 °C) leads to the crystallization of SiO₂, a similar form of cristobalite and phosphate quartz in RHA, thus reducing the chemical activity [26]. The chemical compositions of RHA from different scholars are summarized in Table 1.

The chemical compositions of RHA obtained under different conditions are not exactly the same, but the main component is SiO₂. Generally, there is a small amount of K₂O in RHA, which increases the alkalinity of pore solution of cement pastes. It can

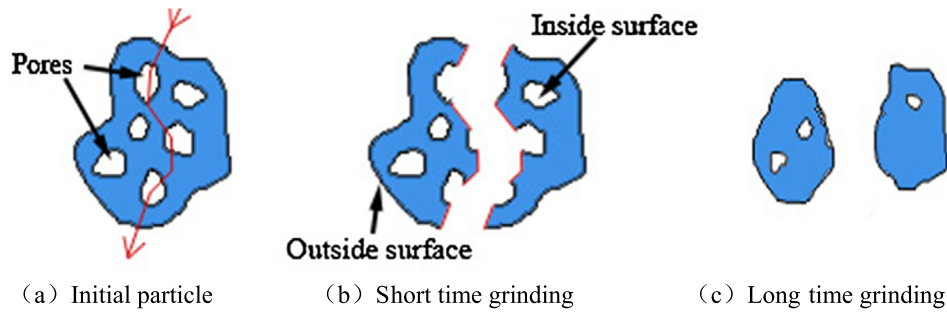


Fig. 3. Variation of RHA particles state during grinding [15].

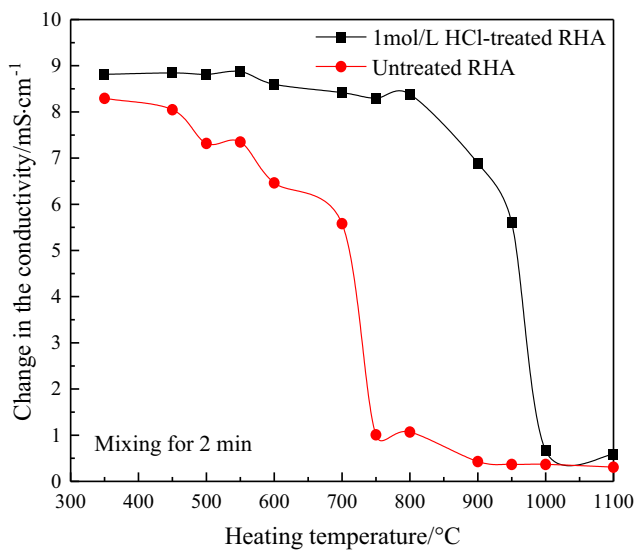


Fig. 4. The relationship between the calcination temperature of RHA and the variation values of saturated CH solution conductivity [37].

not only promote the dissolution of SiO_2 particles, but also help to alleviate the decrease of pore solution pH caused by the pozzolanic reaction between SiO_2 and CH [19]. In ASTM C150/C150M-18 [34], the equivalent alkalis ($\text{Na}_2\text{O} + 0.658\text{K}_2\text{O}$) in cement is recommended to be less than 0.6%. It can be seen from Table 1 that the equivalent alkalis of most RHAs are higher than 0.6%, which is at risk of inducing alkali aggregate reaction. While the K_2O content can be regulated by pretreating the rice husk and changing the combustion conditions [14]. In addition, under the premise of ensuring that the alkali content is less than 0.6%, it is necessary to further study the effect of K_2O content on the pore solution pH and pozzolanic reaction of SiO_2 . The best application range of K_2O content obtained can provide a relevant basis for better application of RHA in cement-based materials.

2.3. Characterization and evaluation method of RHA

1) Porosity characterization of RHA particles

In order to characterize the porosity of RHA particles, the water absorption of RHA was performed by Kang [17]. 0.3 g of RHA dried at $110 \pm 5^\circ\text{C}$ was placed on an electronic balance, at constant temperature and humidity of 20°C and 100% RH, to measure the mass variation of RHA with time. Fig. 2 shows the change of water absorption mass (by unit mass of RHA) with time.

The water absorption behavior of all kinds of RHA tends to be stable after 12 h, the moisture absorption capacity of white RHA is stronger than that of black RHA which is not fully burned. It may be due to the incomplete combustion of organic matter which leads to the incomplete separation of carbon and occupies part of the pores. After grinding, part of the granular pore structure of RHA is destroyed. As shown in Fig. 2, the water absorption capacity of RHA after grinding is lower than that before grinding, which may be due to the collapse of part of the porous microstructure of RHA [35]. The change of porous microstructure of RHA particles during the grinding process is shown in Fig. 3. The water absorption capacity of RHA directly determines the water content provided by RHA in the later-stage hydration of cement. Therefore, the quantitative relationship between the water absorption capacity and internal curing effect of RHA needs to be further studied.

2) Pozzolanic activity evaluation of RHA

The pozzolanic activity of RHA can be evaluated by conductivity measurement [36], which is based on the conductivity change of saturated CH solution. When the pozzolanic materials are added to the saturated CH solution, the active component reacts with CH, resulting in a decrease of Ca^{2+} concentration, thus reducing the conductivity of the solution. The higher reduction in conductivity of saturated CH solution with added RHA is, the stronger the activity of RHA is. Fig. 4 presents conductivity variation of the saturated CH solution within the first 2 min with the calcination temperature of two kinds of RHA, i.e. pretreatment by hydrochloric acid and untreated rice husk.

The greater the change values of conductivity, the higher the activity of RHA obtained at the corresponding combustion temperature. Moreover, the activity of RHA after acid treatment is higher than that of untreated RHA. RHA shows much lower pozzolanic activity with the combustion temperature exceeding 750°C . Table 2 shows the effect of the average particle size of RHA on the conductivity of saturated CH solution. It can be seen that the activity of RHA increases with the decrease of its particle size, but the change is not obvious when the average particle size is less than $5.6\ \mu\text{m}$. It

Table 2
Variation of saturated CH solution conductivity [15].

	RHA with different mean particle sizes			
	15.0 μm	7.3 μm	5.6 μm	3.6 μm
Variation in the electrical conductivity (mS/cm)	0.67	0.91	1.05	1.08

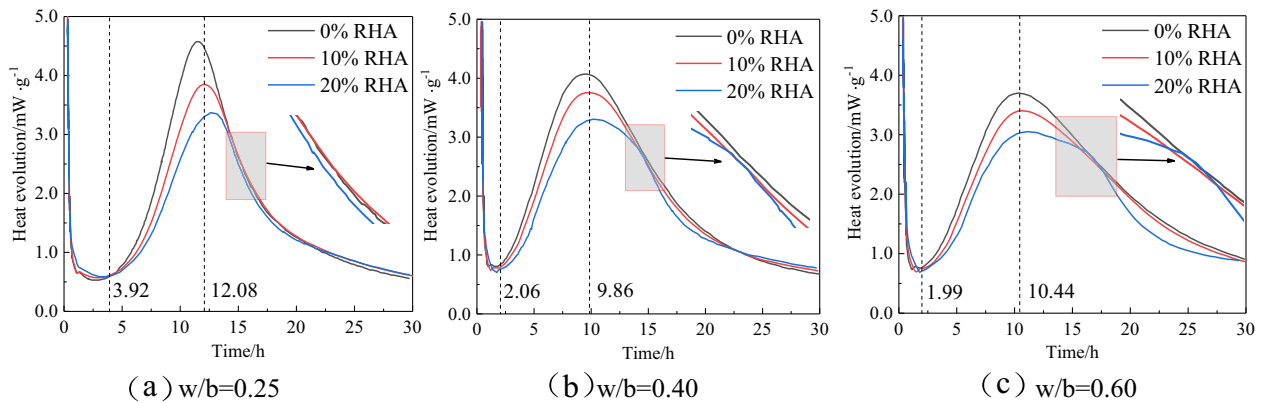


Fig. 5. Effect of RHA content and water-binder ratio on hydration heat of cement [15].

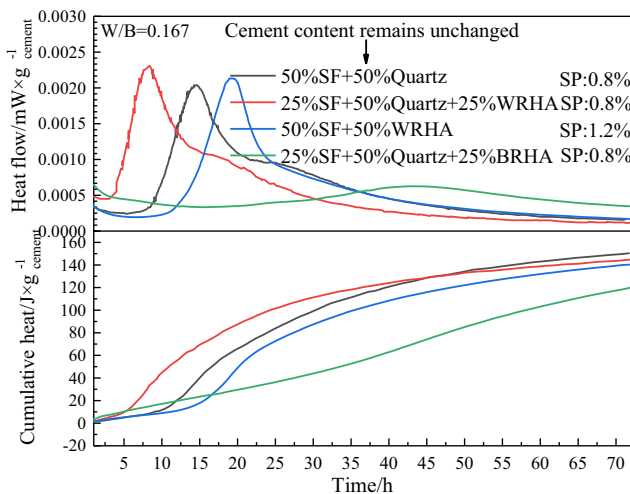


Fig. 6. Effect of RHA on hydration heat of cement [17].

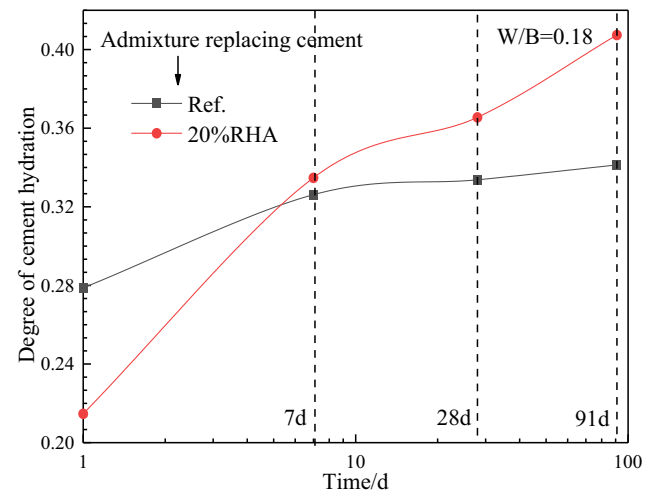


Fig. 7. Effect of RHA on hydration degree of cement [9].

may be due to the increase of grinding degree, which destroys some pores structure of RHA particles, and the porous microstructure collapses and raw particles partially break into finer ones.

3. Hydration properties of cement blended with RHA

The hydration process of blends of RHA and cement, compared with conventional pozzolanic materials, is more complicated due to the adsorption and release of water by the porous structure of RHA. The effects of RHA on cement hydration include dilution effect, chemical effect and internal curing effect [38]. The dilution effect represents the decrease of cement content. The chemical effect is the pozzolanic reaction of SiO_2 with CH [39,40], and the internal curing effect shows the promotion of cement hydration by the continuous release of water from RHA. That is, when the relative humidity in the matrix decreases, water is released from the pores of RHA particles and promotes the further hydration of the surrounding cement particles [41].

3.1. Hydration process of cement blended with RHA

The effect of RHA on the cement hydration is closely related to the water-binder ratio and the content of RHA [15]. With the increase of RHA content, the amount of cement in the pastes decreases and the main hydration exothermic peak is delayed with fixed w/b ratio. When the RHA content is constant, with higher w/b

the main hydration exothermic peak and acceleration period appear slightly earlier. When the w/b are 0.4 and 0.6 respectively, the extra peaks of cement pastes with 20% RHA appear during the hydration deceleration period, and the peak is more obvious when the w/b is 0.6, which may be due to the promoting effect of water released by RHA on cement hydration or pozzolanic reaction of RHA, as shown in Fig. 5. It still need to be further confirmed.

In the first few minutes of cement hydration, Ca^{2+} and OH^- ions are adsorbed on the surface of RHA particles. OH^- increases the coordination number of silicon atoms on the surface and weakens the silicon-oxygen bond. The silicon hydroxyl structural units are formed on the surface. Thereby, silica dissolves in the solution to participate in the reaction. The pozzolanic reaction of RHA affects the precipitation of CH and the formation of C-S-H gel, which further affects the hydration rate of cement. The increasing RHA content reduces the amount of water available for cement hydration, affects the transfer of Ca^{2+} from the particle surface to the solution, delays the hydration of cement, and reduces the OH^- content on the surface of RHA particles. OH^- may not reach the concentration that RHA begins to dissolve in the first few minutes of hydration, thus reducing the dissolved amount of silica and further affects the hydration rate of cement.

The pozzolanic reaction between RHA and CH forms C-S-H(I) -type fibrous gel [42], and some researches reported the formation of fine C-S-H gel ($\text{Ca}_{1.5}\text{SiO}_{3.5}\cdot x\text{H}_2\text{O}$) [43]. C_3S and C_2S in cement par-

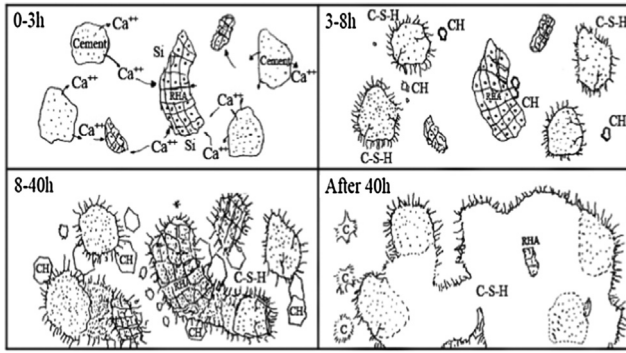


Fig. 8. Mechanism of pozzolanic reaction of RHA [50].

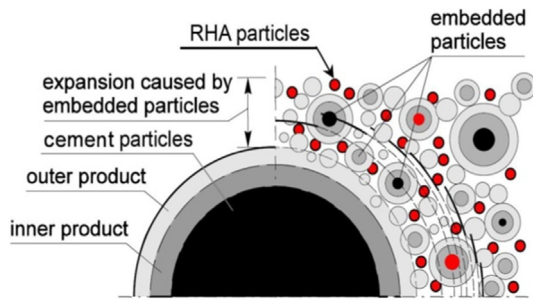
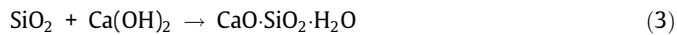
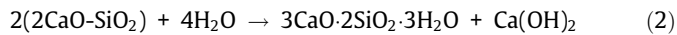
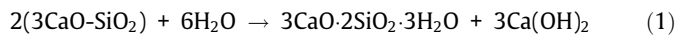


Fig. 9. Shell layer of hydration products of cement and RHA particles [15].

ticles hydrate to form C-S-H gel and CH, which can be expressed as Eq. (1) and Eq. (2). CH reacts with active SiO_2 in RHA to form new C-S-H gel [44], as seen in Eq. (3).



Both RHA and silica fume have high hydration activity, while RHA also has an internal curing effect. By comparing the effects of silica fume and RHA on cement hydration, the hydration reaction of RHA can be understood more clearly. Fig. 6 compares the effects of white RHA and black RHA on the heat evolution curve of cement hydration. Compared with the reference sample, 25% white RHA significantly accelerated the hydration reaction, the main hydration exothermic peak was advanced and the induction period was shortened. The lower the effective water-cement ratio of cement paste is, the faster the formation of the main peak is, and the main hydration reaction starts in advance [45–47]. The absorption of water by RHA reduces the effective w/c of cement paste, and cement hydration enters the acceleration period ahead of time. Feng [37] obtained the similar conclusion in the hydration heat evolution of cement-RHA system, however, different from the conclusion obtained in Fig. 5. Fig. 5 shows that the cement is partial replaced by RHA, i.e. the internal-addition method. In Fig. 6 the external-addition method is adopted to keep the amount of cement unchanged. However, due to the change of cement content and the higher water demand of RHA, the different incorporation methods have different effects on the heat evolution of cement hydration.

As seen in Fig. 6, 50% white RHA delayed the formation of the main exothermic peak of cement paste, prolonged the induction period, slowed down the hydration reaction and reduced the cumulative hydration heat. Although the increase of RHA reduces

the effective water-cement ratio and accelerates the hydration, but the delay of hydration reaction may also be attributed to the increase of superplasticizer, i.e. the action of steric hindrance between cement particles [48,49]. The incorporation of 25% black RHA significantly reduced the hydration rate of the acceleration period and the cumulative hydration heat for 72 h, which may be due to the negative effect of higher carbon content in black RHA on cement hydration. Therefore, it is proved that the carbon content in RHA has a great influence on the hydration reaction of cement.

Fig. 7 shows the effect of RHA addition on the hydration degree of cement. The hydration degree of cement mixed with RHA increased significantly after 7 days, and was higher than that of pure cement. Due to the water absorption of the pore structure of RHA particles, there is less water available for cement reaction in the system, and the early hydration degree of cement is low. With the hydration proceeding, the internal relative humidity of cement paste decreases, the released water from RHA particles promotes the hydration of cement. Meanwhile, the released water also promotes the pozzolanic reaction between Ca^{2+} and SiO_2 in cell space of RHA [50,51]. In addition, the low Ca^{2+} concentration caused by RHA pozzolanic reaction also promotes the hydration of cement.

3.2. Hydration mechanism and model of cement-RHA system

In the hydration process of cement, the water absorbed by the structure of pores makes Ca^{2+} accumulate on the surface of RHA particles and form CH. After 40 h, amorphous SiO_2 reacts with CH and forms more C-S-H gel, and the hydration mechanism is shown in Fig. 8. The initial reaction between RHA particles and CH is a boundary reaction. With the formation of the shell of secondary hydration products, the reaction mechanism changes from boundary reaction to diffusion-controlled reaction, which is similar to the hydration process of cement particles, as shown in Fig. 9.

The hydration product shell around RHA particles is controlled by the pozzolanic reaction. The compactness and porosity of shell as well as the diffusion-controlled reaction are different from that of cement particles. RHA particle is wrap with a water film, which further affects the pozzolanic reaction and diffusion-controlled reaction of RHA as well as the diffusion reaction of cement particles. In the blends of cement and RHA, the hydration involves the reaction mechanism of cement particles and RHA particles. The reaction of cement restricts the reaction of RHA, and the redistribution of water from RHA promotes the reaction of the former.

The hydration reaction of cement-RHA system is complicated. Based on the hydration mechanism, the pozzolanic reaction of RHA is not only affected by chemical activity and physical properties (porosity) of the particles, but also restricted by the hydration reaction of cement. Some scholars put forward the hydration model of RHA from different views. Feng [37] et al argued that the reaction kinetics of RHA with CH is consistent with the diffusion control, which can be characterized by Jander diffusion equation [52], as seen in Eq. (4):

$$F(G) = \left[1 - (1 - G)^{1/3}\right]^2 = 2kt/r^2 = Kt \quad (4)$$

where G and r denote the mass fraction of the reacted CH and the initial radius of the particles. The coefficient k is the parabolic rate constant, and K is the pozzolanic reaction rate constant. The diffusion equation only characterizes the pozzolanic reaction of RHA in saturated CH solution, which can not accurately reflect the hydration reaction of RHA in complex cement system. Nguyen [15] adopted the modified HYMOSTRUC3D composite cement hydration model to simulate the hydration process of cement-RHA system. Similar to the hydration kinetics of cement, the initial

reaction rate of RHA particles is also determined by the phase boundary reaction. When the shell of the hydration product reaches a certain thickness, the phase boundary reaction transforms into a diffusion-controlled process, and the basic rate equation is shown in Eq. (5):

$$\frac{\Delta \delta_{in,xj+1}^{RHA}}{\Delta t_{j+1}} = K_0^{RHA} \times \Omega_1 \times \Omega_2 \times \Omega_3 \times F_1^{RHA} \times \left[F_2^{RHA} \times \left(\frac{\delta_{tr}^{RHA}}{\delta_{xj}^{RHA}} \right)^{\beta_1} \right]^{\lambda} \times P_{ORHA} \quad (5)$$

where K_0^{RHA} and F_1^{RHA} denote the initial reaction rate constant of RHA and the function of the effect of temperature on hydration rate of RHA, and F_2^{RHA} is the temperature function of the effect of curing temperature on the structure in the diffusion control step. The reduction coefficient of water absorption of central RHA particles is represented by Ω_1 , which is the reduction of permeability rate of the particle shell caused by the unhydrated RHA particles embedded in the central particle shell for further water absorption. Ω_2 is the reduction coefficient of water shortage of pores caused by reaction, it can be deduced from the volume of capillary water and the maximum pore size in the system [53]. Ω_3 is the reduction coefficient of the amount of water involved in the hydration reaction of cement and RHA. λ is the coefficient of rate control, which is 0 for the phase boundary reaction and 1 for the diffusion control process. δ_{tr}^{RHA} is the thickness of the transition layer of RHA particles, i.e. the thickness of the hydration products layer when the phase boundary reaction is transformed into the diffusion control reaction. δ_{xj}^{RHA} is the shell thickness of hydration products of arbitrary RHA particles, which is determined by the hydration degree of RHA particles. β_1 is an empirical constant. P_{ORHA} is the reaction coefficient of RHA and CH, the value is 1 when the content of CH is enough, and the value is the effective CH content when the content of RHA is insufficient.

Based on the modified hydration model mentioned above, Nguyen [15] simulated the effect of RHA particle size and content on cement hydration, which was in good agreement with the experimental data hydrating for 100 h. The results showed that the finer the RHA particles are, the faster the pozzolanic reaction is and the more C-S-H are formed. The content of RHA to replace cement is recommended to be less than 30%. However, the internal curing effect of RHA, i.e. the water absorption and release in porous RHA particles, was not taken into account in the modified hydration model. Park [38] put forward the hydration model of cement-RHA system by combining the factors, e.g. the adsorbed water, the water-binder ratio, the content and fineness of RHA, the content of amorphous SiO_2 , etc. As shown in Eq. (6) to Eq. (14).

According to the stoichiometric method of amorphous SiO_2 reaction proposed by Bentz [54], the CH content is calculated, as seen in Eq. (6):

$$CH = RCH_{CE} \times C_0 \times \alpha - 1.36 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s \quad (6)$$

where RCH_{CE} and C_0 denote the mass of CH (g) produced by hydration of unit mass cement and the mass fraction of cement respectively. α is the hydration degree of cement. α_{RHA} is expressed as the hydration degree of glass phase in RHA. m_{RHA0} represents the mass of RHA in mixture. γ_s is the mass fraction of amorphous silica in RHA. $RCH_{CE} \times C_0 \times \alpha$ is the CH produced by cement hydration, and $1.36 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s$ is the CH content consumed by secondary hydration reaction of RHA. For the pozzolanic reaction of amorphous SiO_2 in conventional pozzolanic materials, 1 g amorphous SiO_2 consumes 0.5 g gel water [55]. The capillary water and chemical combined water can be calculated by Eqs. (7) and (8):

$$W_{cap} = W_0 - 0.4 \times C_0 \times \alpha - 0.5 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s \quad (7)$$

$$W_{chem} = 0.25 \times C_0 \times \alpha \quad (8)$$

where W_{cap} and W_{chem} denote capillary water and chemically combined water respectively. W_0 is the mass fraction of water in the mixture. $0.4 \times C_0 \times \alpha$ represents the capillary water consumed by cement hydration. $0.5 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s$ represents the capillary water consumed by amorphous SiO_2 pozzolanic reaction. The chemically bound water of 1 g completely hydrated cement is about 0.23 g [55], while 0.25 g for CEM I [56]. Due to the redistribution of water in the porous structure of RHA during cement hydration, it is necessary to consider the adsorption and release of capillary water by RHA particles in the calculation of the model, as shown in Eqs. (9)–(11),

$$W_{cap} = W_0 - (m_{RHA0}/\rho_{RHA}) \times \varphi_{RHA}, \quad (t = 0) \quad (9)$$

$$W_{cap} = W_0 - 0.4 \times C_0 \times \alpha - 0.5 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s (m_{RHA0}/\rho_{RHA}) \times \varphi_{RHA} \leq 0.0625 \times C_0 \times \alpha + 0.22 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s \quad (10)$$

$$W_{cap} = W_0 - 0.4 \times C_0 \times \alpha - 0.5 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s - [(m_{RHA0}/\rho_{RHA}) \times \varphi_{RHA} - 0.0625 \times C_0 \times \alpha - 0.22 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s (m_{RHA0}/\rho_{RHA}) \times \varphi_{RHA}] \geq 0.0625 \times C_0 \times \alpha + 0.22 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s \quad (11)$$

in which $t = 0$ is the time when the mixture is in contact with water. φ_{RHA} is the porosity of RHA. $(m_{RHA0}/\rho_{RHA}) \times \varphi_{RHA}$ considers the absorption of water by the porous structure of RHA. $0.0625 \times C_0 \times \alpha$ is the water absorbed from RHA because of the chemical shrinkage caused by cement hydration. $0.22 \times \alpha_{RHA} \times m_{RHA0} \times \gamma_s$ is the water absorbed from RHA because of the chemical shrinkage caused by the secondary hydration of RHA [57]. When the water absorbed by RHA is less than the chemical shrinkage of the mixture, then water content is enough for continued hydration reaction. The relative humidity in the cementitious system is relatively high, which makes it difficult for RHA to release water without the humidity gradient. Hence, it can be approximately considered that the released water is 0. When the water absorption of RHA is more than the chemical shrinkage of the mixture, the water in the pores of RHA is released, and the capillary water content is calculated by Eq. (9).

Based on the formation and destruction rate of initial hydration layer, active chemical reaction process and diffusion-controlled process in cement hydration model [58,59], Park [38] proposed the pozzolanic reaction of RHA as follows: initial dormancy process, phase boundary reaction process and diffusion process. Saeki [60] further deduced the hydration model of RHA, as seen in Eq. (12)–(14):

$$\frac{d_{zRHA}}{dt} = \frac{m_{CH}(t)}{m_{RHA0}} \frac{3\rho_w}{v_{RHA}r_{RHA0}\rho_{RHA}} \times \frac{1}{\left(\frac{1}{k_{dRHA}} - \frac{r_{RHA0}}{D_{eRHA}} + \frac{r_{RHA0}}{D_{eRHA}} (1 - \alpha_{RHA})^{-1/3} + \frac{1}{k_{rRHA}} (1 - \alpha_{RHA})^{-2/3} \right)} \quad (12)$$

$$k_{dRHA} = \frac{B_{RHA}}{(\alpha_{RHA})^{1.5}} + C_{RHA} \times (\alpha_{RHA})^3 \quad (13)$$

$$D_{eRHA} = D_{eORHA} \times \ln \left(\frac{1}{\alpha_{RHA}} \right) \quad (14)$$

where $m_{CH}(t)$ denotes the CH content of per unit volume at a certain time. v_{RHA} is the stoichiometric ratio of CH to RHA, and ρ_w is the density of water. r_{RHA0} is the radius of RHA particles, $r_{RHA0} = 3/(S_r \rho_r)$. S_r and ρ_r denote the specific surface area and density of RHA. k_{dRHA} is the reaction rate coefficient of RHA on the initial

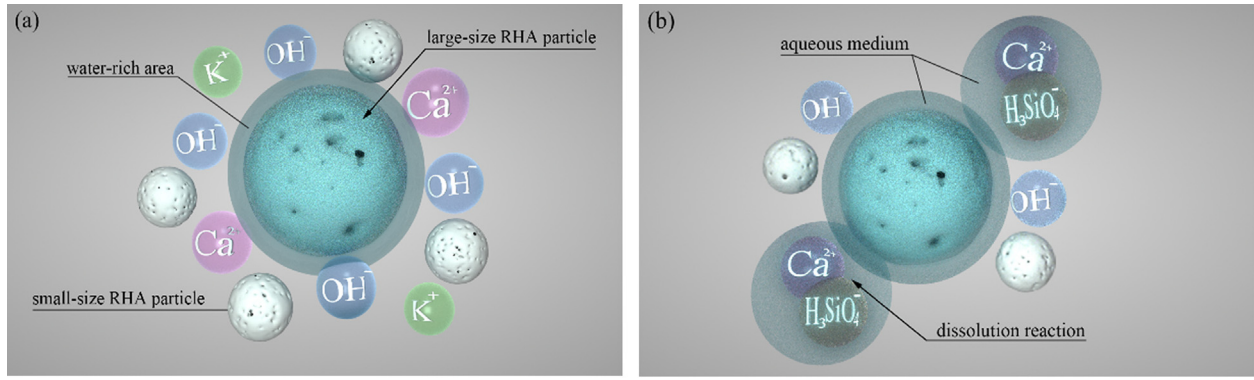


Fig. 10. Dissolution reaction diagram of RHA particles.

dormancy period. k_{rRHA} is the reaction rate coefficient of RHA in the phase boundary reaction process. B_{RHA} and C_{RHA} are the formation rate and destruction rate of the initial hydration layer, respectively. D_{eRHA} and D_{eORHA} represent diffusion coefficient of water adsorbed by RHA and the initial diffusion coefficient.

Park [38] approximately simulated the absorption and release of water by RHA in the cementitious system. While the release of adsorbed water is not only related to the chemical shrinkage of cementitious materials and the total pore volume of RHA, but also related to the saturation degree of RHA, the sorption and desorption of RHA, the relative humidity gradient between RHA particles and the surrounding hydration products, the pore size distribution of RHA, the redistribution of local moisture and the mass and the momentum conservation of the liquid moisture and the vapor [61]. Hence, in order to accurately analyze the hydration process of cement-RHA system and the role of RHA characteristics in the hydration process, it is necessary to combine more factors for further research.

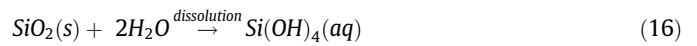
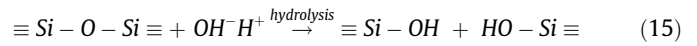
3.3. Discussion on hydration of RHA particles

Based on the analysis and consideration of physicochemical effects of RHA in cement-based materials, two hydration reaction modes of RHA particles are summarized based on the porous structure characteristic and particle size of RHA.

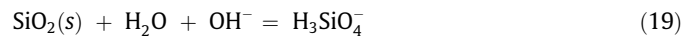
3.3.1. Dissolution reaction of micro-particles

A large amount of OH^- are released in the early-stage of cement hydration, which make the pastes solution alkaline. At high pH, the polar siloxane bonds (Si-O) in SiO_2 are broken by the nucleophilic attack of hydroxyl ions (OH^-) [62]. As a result, the SiO_2 monomer

particles dissolve in the pastes solution and then hydrolyze to form large amount of $\equiv Si-OH$ groups, as seen in Eq. (15) and Eq. (16).



When pH greater than 10, the solubility of SiO_2 particles becomes more significant [63], and the rate of dissolution increases linearly with pH [64]. The $Si(OH)_4$ monomers produced by hydrolysis undergo ionization to form $H_3SiO_4^-$ and $H_2SiO_4^{2-}$, as seen in Eq. (17) and Eq. (18). It can also be expressed as Eq. (19) by combining Eq. (16) [65].



After SiO_2 dissolve in alkaline condition, the main phases in aqueous solution are $Si(OH)_4$ monomers, $H_3SiO_4^-$ and $H_2SiO_4^{2-}$ [66]. The free Ca^{2+} react with these monomers in the solution to form low Ca/Si C-S-H gels. As known, RHA has large amount of amorphous silica. Assuming that the RHA particles are spherical, and combining with the reaction mechanism mentioned above, the dissolution reaction diagram of RHA particles is expressed in Fig. 10.

In Fig. 10(a), the porous structure on the surface of RHA particles can absorb a certain amount of water and form a water-rich layer, which is beneficial to the adsorption of dissolved Ca^{2+} and

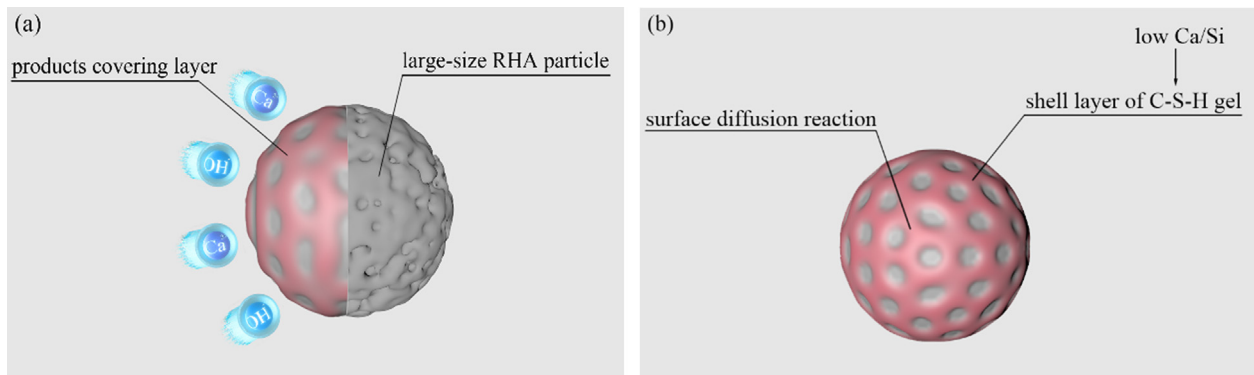


Fig. 11. Diffusion reaction diagram of RHA particles.

OH^- . The K_2O in RHA can be hydrolyzed rapidly and ionized out OH^- , which further increases pH of the solution and promotes the dissolution of RHA particles. However, not all particles will dissolve, and the small-size particles dissolve preferentially to that of large ones. The dissolved H_3SiO_4 interacts and reacts with the adsorbed Ca^{2+} , as shown in Fig. 10(b). Due to the consumption of OH^- by the dissolution of partial RHA particles reduces the pH of the surrounding solution environment, which will slow down the dissolution of other micro-particles. Therefore, in the initial stage of cement hydration, the pozzolanic reaction of RHA should be the dissolution of partial micro-particles. With the development of hydration, the increase of OH^- will make other micro-particles dissolve slowly and react.

3.3.2. Diffusion reaction of macro-particles

The large-size RHA particles are difficult to dissolve, so the reaction mode is different from that of small particles. In the early stage of hydration, the large particles are wrapped by water film. Under the action of liquid-phase mass transfer, Ca^{2+} and OH^- are enriched to the liquid-solid interface and the phase boundary reaction occurs when interacting with SiO_2 molecules on the surface of large particles. It takes the surface of the particle as the nucleation sites, and forms a product covering layer, as shown in Fig. 11(a). With the processing of hydration, Ca^{2+} and OH^- diffuse and react on the surface of large particles until the hydration product (C-S-H gels) shells wrap the entire particle, as seen in Fig. 11(b). In fact, the mesoporous structure of large particles absorbs large amount of water, and Ca^{2+} , OH^- in the initial stage can also diffuse to the interior of the pores [67]. Thus, the phase boundary reaction and surface diffusion reaction shown in Fig. 11 also occur on the inner surface of the pores.

When the product shell reaches a certain thickness so that Ca^{2+} and OH^- cannot directly contact with the particle surface, the reaction mode of large particles transforms into diffusion reaction of ion penetrating surface shell to internal mass transfer, as the diffusion-controlled reaction mentioned in the previous Nguyen [15] hydration model and shown in Fig. 9. The diffusion reaction rate of large particles is controlled by the compactness and pores of the product shell, and its hydration reaction is carried out slowly. Therefore, in the whole process of cement hydration, the contribution of large-size RHA particles to the system is more inclined to the release of water caused by its internal curing effect.

3.3.3. Summary

According to the two hydration modes mentioned above, the particle size is one of the main factors affecting the hydration of RHA. The dissolution reaction of small-size particles occurs in high alkali environment, and the diffusion reaction of large-size particles occurs slowly after the surface phase boundary reaction. The C-S-H gels produced by dissolution of small particles can fill the pores of the cement matrix, and the product shell on the surface of large particles will increase a certain thickness, and the hydration products grow outward to inward. Therefore, it is reasonable to consider that the contribution of small-size particles to the system represent the chemical effect of RHA, and the large-size particles represent the internal curing effect of RHA. So, how to delimit the large or small of RHA particles? At present, this is still a blank in the research on application of RHA. If the “large” and “small” size mentioned above or size ranges of RHA particles can be discriminated in the further research, the quantitative distinction between the chemical effect and internal curing effect of RHA can be achieved. Furthermore, the proportion of the two RHA effects can be controlled by grinding technology, so as to satisfy the specific requirements of different engineering for cement-based materials.

4. Effect of RHA on the performance of cementitious system

RHA continuously releases water with the change of relative humidity during the formation and microstructure evolution of cement matrix. It promotes the further hydration of cement and the pozzolanic reaction of RHA, forming more C-S-H gel with low Ca/Si ratio, filling the pores of cement pastes, and reducing porosity. In terms of macro-properties, the contributions of RHA to the cementitious system are mainly reflected in the increase of strength [68–72] and the improvement of durability [73–75], e.g. chloride penetration resistance and sulfate resistance, etc.

4.1. Strength

Sufficient burned RHA contains a large amount of active SiO_2 under certain conditions. RHA has the characteristics of high specific surface area and porous structure, which have many effects on the strength of cement matrix. It can provide nucleation sites for cement hydration, thus promote the hydration reaction of cement. Moreover, the fine particles of RHA can fill the pores of matrix and its pozzolanic reaction can form additional C-S-H gel to further fill

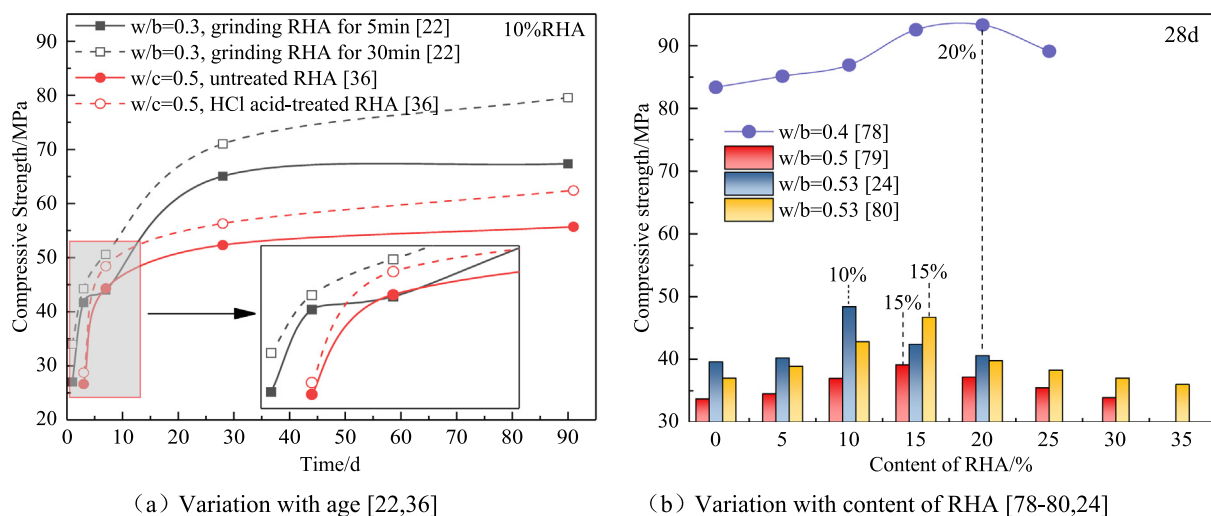


Fig. 12. Effect of RHA on compressive strength of cement-based materials.

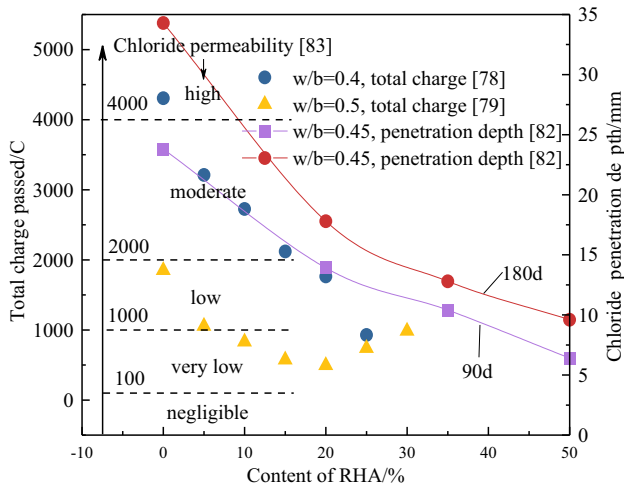


Fig. 13. Effect of RHA on chloride permeability of cement-based materials [79,80,83,84].

the pores and increase the compactness of the matrix. In addition, the release of water from the porous structure of RHA promotes the further hydration of cement. Therefore, the compressive strength and its development of cement matrix can be effectively improved under the physical and chemical action of RHA [76,77]. The tensile strength and flexural strength of the cement paste blended with RHA are also increased similarly [23,37,78,79–81].

The variation of compressive strength of cement-RHA cementitious materials with age is summarized in Fig. 12(a). The ground RHA significantly improves the compressive strength of the matrix, and the hydrochloric acid pretreatment can avoid the influence of impurities in RHA. Compared with the untreated RHA, the acidified RHA can effectively improve the matrix strength. Fig. 12(b) shows the change of compressive strength with RHA content. Under different water-binder ratios, the 28 days strength of the matrix increased at first and then decreased with the content of RHA. The optimum contents of RHA for improving strength obtained by different scholars were 10%, 15% and 20%, respectively. With the increase of RHA content, the water absorption also increases, which leads to the decrease of the water available for cement hydration. While the decrease of cement content leads to the decrease of hydration products and the insufficient content of CH

for RHA pozzolanic reaction. The comprehensive effects lead to the reduction of the matrix strength. When more RHAs are added to the mixture, the other particles will be used as inert filling materials except fine particles, which can not promote the development of matrix strength [25].

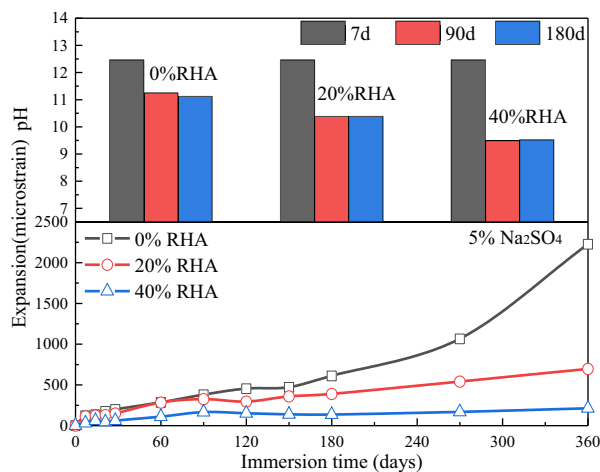
4.2. Durability properties

4.2.1. Chloride penetration resistance

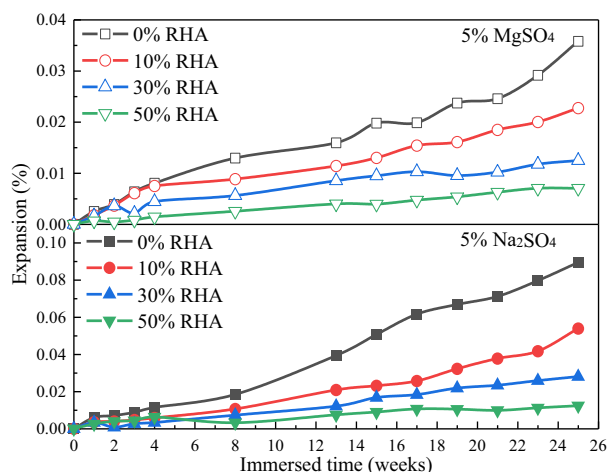
Pozzolanic reaction and internal curing effect of RHA promote the hydration of cement in the later stage, which gradually reduces the porosity of the matrix and improves the resistance to chloride ion penetration of cement-based materials [82,11]. Fig. 13 shows the variation of chloride ion penetration depth and electric flux of cement matrix with the content of RHA.

RHA can effectively improve the chloride ion permeability of concrete, which is 4 times higher than that of normal concrete without RHA [85]. In Fig. 13, the chloride ion penetration depth of samples for 90 days and 180 days decrease gradually with the RHA content. Furthermore, the total passed charge decreases with RHA content linearly, and another result shows that first decreases and then increases with RHA. It was also reported that the increase of RHA may lead to the decrease of electric flux and the chloride ion permeability decreased with the increase of RHA content [86]. The addition of RHA increases the silicate ion in the pore solution, which improves the hindrance effect of cement-based materials on the migration of chloride ion. With the increase of C-S-H, the physical adsorption of chloride ion is improved, and the gradually dense pore structure reduces the diffusion path of chloride ion [83,87]. The reduction of resistance to chloride ion permeability caused by excessive RHA may be due to the decrease of amount of C-S-H, which is attributed to the decrease of cement and the slow pozzolanic reaction because of the increase of water absorption. Nonetheless, the resistance to chloride ion permeability is still higher than that of the reference sample, which indicates that the high content of RHA improves the compactness of the matrix by filling the pores [25].

Similarly, Gastaldini [88] found that the total charge passed of the paste with 20% RHA was reduced by 65%, 68% and 59%, respectively at the w/b of 0.35, 0.5 and 0.65 compared with the samples that without RHA. Ganesan [81] adopted 30% RHA to replace cement, and the chloride ion permeability was reduced by 75% compared with the sample without RHA. The chloride ion permeation test of the sample mixed with RHA was carried out by



(a) Expansion and pH [91]



(b) Expansion in different solutions [92]

Fig. 14. Effect of RHA on sulfate resistance.

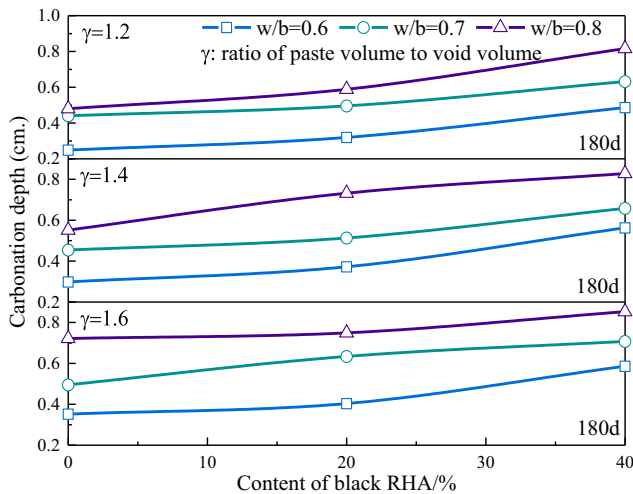


Fig. 15. Effect of RHA on carbonation resistance of concrete [96].

Madandoust [89]. The higher the content of RHA was, the lower the chloride ion concentration was, and the stronger the resistance to chloride ion penetration was.

4.2.2. Sulfate resistance

The resistance to sulfate attack of cement-based materials depends on the composition and microstructure of cementitious materials. Reducing the content of C_3A and CH , and densifying the microstructure of the matrix can raise the resistance to sulfate attack [90]. It is recognized that addition of pozzolan generally improves the durability of concrete especially for the sulfate resistance [91]. The reduction of C_3A by dilution effect of RHA, the consumption of CH by pozzolanic reaction and the improvement of microstructure by internal curing effect can improve the resistance to sulfate attack of cement matrix. The effect of RHA on the sulfate resistance is summarized in Fig. 14.

The sample without RHA showed the accelerating expansion with age after 120 days in Fig. 14(a). The sample mixed with RHA did not show significant accelerating expansion, and the sample mixed with 40% RHA still showed a small expansion after 360 immersed days. As seen in Fig. 14(b), the expansion rate of the samples decreased with the content of RHA after 8 weeks under different sulfate environments. Therefore, it can be seen that the concrete expansion is effectively reduced by the incorporation of RHA, and the sulfate resistance increases with RHA content. Sensale [82] also obtained the same result. The consumption of CH by pozzolanic reaction of RHA significantly reduced the pH of solution. At pH of 12–12.5 only ettringite formation can take place and at pH of 8.0–11.5 gypsum formation and decalcification occur. The decalcification of C-S-H gel with high Ca/Si ratio will form a porous layer, the decalcification of C-S-H gel with low Ca/Si ratio can form a protective layer of silica phase [92], and the pozzolanic reaction of RHA will form a low Ca/Si C-S-H gel. The protective layer formed during the decrease of solution pH prevents sulfate penetration and the further corrosion. However, in the sulfate environment of different cations, Mg^{2+} with 2 valence electrons is easier to substitute for the Ca^{2+} in C-S-H than Na^+ with 1 valence electron, which leads to the transformation of C-S-H structure to M-S-H with lower binding ability [93]. Under sulfate corrosion environment, CH reacts with sulfate to form gypsum, and gypsum will continue to react with aluminate to form ettringite, which results in the expansion and crack of cement matrix. The content of CH can be effectively reduced by increasing the content of RHA, and then decrease the formation of ettringite. Furthermore, the protective

layer of silica phase produced by decalcification of C-S-H with low Ca/Si can prevent the erosion of sulfate. Thus, the resistance to sulfate attack of concrete can be improved by RHA [94,95]. However, the unlimited increase of the RHA content may adversely affect the sulfate resistance or other properties of concrete and this needs to be further researched.

4.2.3. Carbonation resistance

CO_2 can penetrate into the surface of concrete and react with CH in cement paste, which reduces the pH value of pore solution and destroys the protective layer of steel bar. The consumption of CH by RHA and the pozzolanic reaction make the cement matrix form a dense structure and reduce the carbonation of cement-based materials. Fig. 15 shows that the effect of black RHA content with higher carbon content on carbonation depth of concrete on 180 days. The carbonation depths of the samples with RHA are higher than that of the sample without RHA. Under different w/b and different ratios of paste volume to void volume, the carbonation depths increase with the content of RHA. The pozzolanic activity is reduced due to the high carbon content of black RHA, which makes the consumption of CH reduce. The porosity of cement matrix is increased, which leads to the increase of carbonation depth. However, Gastaldini [88] showed that when the content of RHA was 20%, the carbonation depth of concrete could be effectively reduced. When the w/b were 0.5 and 0.65, the carbonization depth of specimen with 20% RHA was 37.47% and 24.47%, respectively, which were lower than that of specimen without RHA. The comparison shows that different qualities of RHA have different effects on the carbonation resistance of concrete. In order to decrease the carbonation depth of concrete, the use of low active RHA with high carbon content should be avoided. At present, there are still few studies about the effect of RHA on carbonation resistance of concrete, and the relevant reliability basis is still not enough, so more and further research needs to be done.

4.2.4. Shrinkage resistance

The volume stability of cement-based materials is directly related to the durability and service life of the structure. The matrix volume will change in the process of cement hardening and curing, and excessive volume deformation will lead to cracking and structural deterioration of cement matrix [97]. When the relative humidity in the matrix decreases, as internal curing material, the water stored of RHA can compensate the relative humidity of the pores with lower humidity. Thus, the capillary pressure at the curved liquid surface of the pore solution decreases, and the volume shrinkage of the matrix reduces. Fig. 16 shows the effect of RHA on the shrinkage resistance of cement-based materials.

In Fig. 16, it can be seen that when the content of RHA is less than 20%, the self-shrinkage deformation of cement matrix decreases with RHA, and the internal relative humidity of the samples mixed with different reactive RHA are higher than that of the sample without RHA after 1 day. Therefore, compared with the internal curing effect, the water release from the porous structure is the main reason for the higher relative humidity of the sample. The pozzolanic reaction of RHA may not consume water, as shown in Eq. (3). Some research [100] has suggested that the impact of pozzolanic reaction on humidity is less. The replacement of cement by RHA leads to a decrease in the amount of hydration cement and a reduction in water consumption, and the compensation of humidity by internal curing effect of RHA make the matrix humidity higher than the reference sample. The higher reaction activity of RHA makes the relative humidity of the sample higher, which may be due to the large pores in the cement matrix [98].

Gastaldini [101] studied the effects of RHA on the shrinkage properties of concrete on 365 days with the water-binder ratios of 0.35, 0.5 and 0.65, respectively. The results showed that the

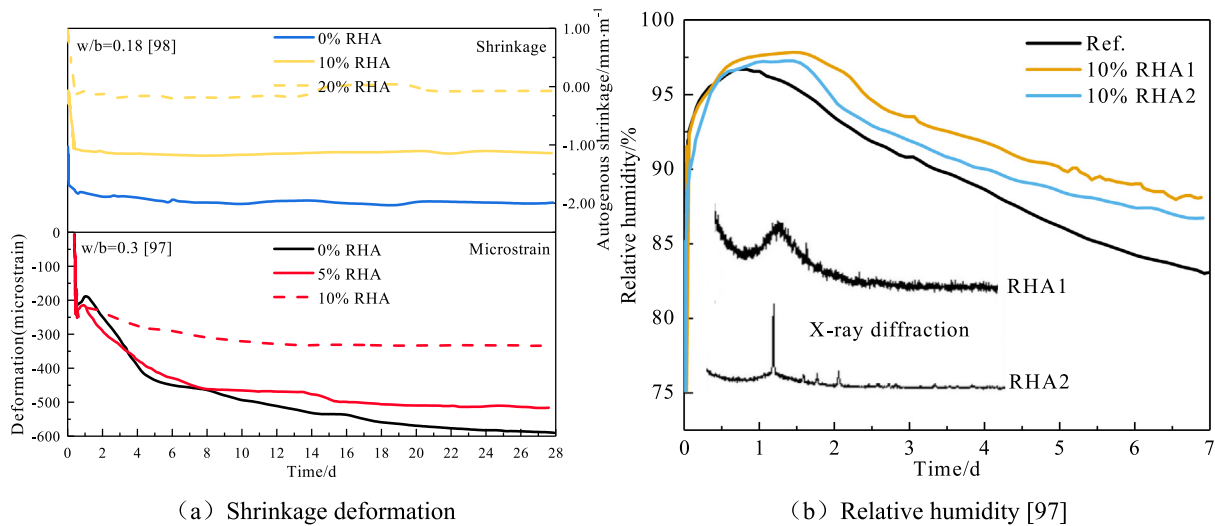


Fig. 16. Effect of RHA on shrinkage and relative humidity of cement-based materials [98,99].

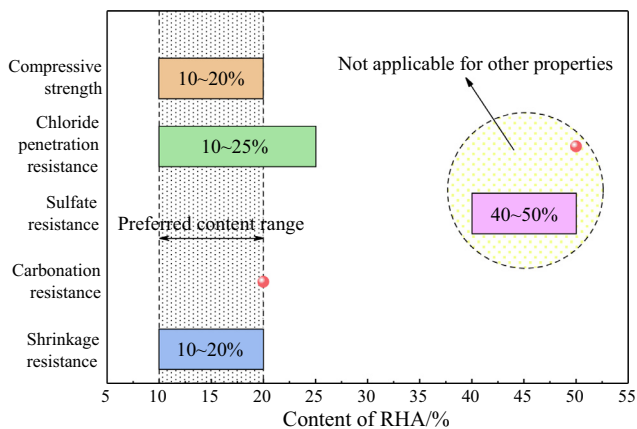


Fig. 17. Preferred content range of RHA used in cementitious materials.

RHA effectively decreases the long-term shrinkage. Although the shrinkage deformation increased with the RHA (5%–30%), however, it was still lower than that of reference sample without RHA. The effects of RHA and SF on the self-shrinkage and relative humidity of UHPC were compared by Van [102]. It shows that RHA has better improvement than SF for UHPC on shrinkage resistance performance. The reduction of relative humidity inside UHPC is significantly delayed due to the internal curing effect of RHA. As a result, the self-shrinkage of UHPC is effectively alleviated. Chatveera [96] found that 20% black RHA with higher carbon content had adverse effect on drying shrinkage of concrete. When the RHA increased to 40%, the drying shrinkage could be reduced properly, which was close to the sample without RHA. The chemical shrinkage caused by pozzolanic reaction was higher than that caused by cement hydration. Highly reactive RHA could make the chemical shrinkage deformation of the sample larger at the early age. However, the shrinkage deformation of the samples mixed with different active RHA remained basically consistent after 16 days. In summary, the improvement of shrinkage resistance of cement-based materials by RHA is mainly attributed to the redistribution of water by its porous structure [98].

4.3. Summary

4.3.1. Preferred content of RHA

Based on summary and analysis mentioned above concerning the effect of RHA on the properties of cementitious system. The optimal content of RHA in different works or water binder ratios are selected. By comparing the optimal content of different properties, the preferred content range of RHA with universality is comprehensively selected, as shown in Fig. 17.

In Fig. 12(b), the RHA content with the highest compressive strength of cementitious materials obtained by different scholars under different water-binder ratios are 10%, 15% and 20%, respectively. With the increase of the RHA content, the strength will decrease gradually. For the chloride penetration resistance in Fig. 13, the optimum content is 25% when the water-binder ratio is 0.4, 50% when the water-binder ratio is 0.45, and the chloride ion permeability stay very low when the water-binder ratio is 0.5 and the content increases from 10% to 25%. The optimal content of sulfate resistance obtained by different scholars in Fig. 14 are 40% and 50%, respectively. The regularities are the same, i.e. the sulfate resistance increases with the content of RHA. Although the increase of RHA content can effectively improve the sulfate resistance, the blindly increase of the RHA content may have adverse effects on other macro-properties. There are few studies on the carbonation resistance of cementitious materials blended with RHA, and 20% RHA can effectively reduce the carbonation of cementitious materials. The optimal content of shrinkage resistance with different water-binder ratios in Fig. 16(a) are 10% and 20%, respectively. By summarizing the effect of RHA on performances, it is considered that the preferred content range of RHA should be 10%–20%, and this content range has good universality to the main performances.

4.3.2. Quality of RHA

The quality of RHA mainly depends on the content of carbon and amorphous SiO_2 . From the previous description, it can be seen that the black RHA, i.e. high carbon content, applies an adverse effect on partial properties of the cementitious system. The incomplete combustion carbon wraps SiO_2 particles and reduces the exertion of activity, which leads to lower pozzolanic activity of RHA. The content of amorphous SiO_2 determines the chemical reaction degree of RHA. Acid leaching pretreatment before calcination is the key for RHA to obtain high content of amorphous SiO_2 ,

because the acid treatment can remove the crystallization catalyst from rice husk and reduce the crystallinity of SiO_2 during combustion. As seen in Fig. 12(a), the acid-pretreated RHA is more beneficial to the improvement of strength of cementitious materials. In summary, the RHA with the lower carbon content and the higher content of amorphous SiO_2 will be more beneficial to the properties of cement-based materials.

5. Conclusions

Based on the results and discussions of the above review, the following conclusions can be obtained.

1. The effect of RHA on cement hydration process is not only related to the content and w/b, but also to the incorporation method. When RHA replaces cement and w/b keeps invariant, with the increase of RHA content, the amount of cement in the pastes decreases and the main hydration exothermic peak is delayed. When the RHA content is constant, the increase of w/b makes the main hydration exothermic peak and acceleration period slightly earlier. The external-addition of RHA can significantly accelerate the hydration reaction of cement, advance the main exothermic peak of hydration and shorten the induction period. But when the content of RHA is as high as 50%, the main exothermic peak is delayed, the induction period is prolonged, the hydration reaction slows down and the cumulative hydration heat is relatively reduced. The high carbon content in RHA has a negative effect on cement hydration, which presents as the obvious reduction of cement hydration rate and the retardation of hydration process.

2. The hydration of RHA particles can be divided into two main reactions: (1) dissolution reaction of micro-particles and (2) diffusion reaction of macro-particles. The dissolution of micro-particles is rapid, and the C-S-H gels produced can fill the pores of the cement matrix. The diffusion reaction of macro-particles develops slowly, and beside that the product shell on surface will increase a certain thickness, the hydration products grow outward to inward. In the process of diffusion reaction, the initial phase boundary reaction on surface is controlled by pozzolanic reaction, which depends on the hydration rate of cement and the diffusion rate of ions. The following diffusion reaction of large particles depends on the packing density and porosity of hydration product shells on the surface of particles. The contribution of small-size particles to the system presents as the chemical effect of RHA, and the contribution of large-size particles presents as the internal curing effect of RHA. The hydration mechanism of cement-RHA system involves the reaction of cement particles and RHA particles respectively, which restrict each other. The hydration reaction of cement restricts the pozzolanic reaction of RHA, and the redistribution of water from RHA promotes the hydration of cement.

3. RHA particles fill pores, pozzolanic reaction forms C-S-H gel to fill pores and the internal curing effect of RHA synthetically increase the compactness of cement matrix, then improve the strength of cement-based materials and promote the development of strength. With the increase of RHA, the compressive strength increases at first and then decreases, and the optimum content of RHA for the improvement of compressive strength is from 10% to 20%. Compared with the untreated RHA, the grinding or acidified RHA can effectively improve the strength of the cementitious material.

4. The improvement of pore structure by RHA and the promotion of cement hydration by internal curing effect reduce the porosity of the matrix and improve the chloride penetration resistance to a certain extent. The increase of RHA reduces the content of C_3A and CH , then reduces the formation of ettringite and improves the sulfate resistance. The low Ca/Si C-S-H gel produced

by pozzolanic reaction of RHA further impedes the sulfate attack. The sulfate resistance increases with increasing amount of RHA, from 10% to 50%. When RHA is 20%, the carbonation of cementitious material is effectively reduced. The water released by RHA in later stage reduces the capillary pressure of cementitious material, which reduces the volume shrinkage of cement matrix. The internal curing effect of RHA plays an important role in improving the shrinkage resistance performance.

5. The effects of RHA content on strength, chloride penetration, sulfate attack, carbonation and shrinkage of cementitious materials are comprehensively investigated. It is considered that the preferred content range of RHA should be 10%–20%, and this content range has good universality to performances. Based on the analysis of RHA affecting the macro-properties mentioned above, the quality of RHA mainly depends on the content of carbon and amorphous SiO_2 . The lower the carbon content and the higher the amorphous SiO_2 content of RHA are, the more beneficial to the properties of cement-based materials is.

6. Recommendation and prospect

1. The quantitative relationship between chemical action and internal curing effect of RHA need to be established. The more pores of RHA, the more water released at the later age, and the more significant the internal curing effect is. In the process of grinding, RHA will lead to the destruction of particle pore structure and reduce the water storage capacity. However, the ground RHA has higher chemical activity in pozzolanic reaction, the refinement of RHA particles improves its own chemical effect and reduces the internal curing effect. As mentioned above, the division between the “large” and “small” size of RHA particles can realize the quantitative distinction between the chemical effect and internal curing effect of RHA. To clarify the contribution degree, pros and cons of chemical effect and internal curing effect of RHA with different fineness to the development of concrete performances is very important for the admixture design of cement and concrete mixed with RHA.

2. The hydration process and mechanism of cement-RHA system need to be further studied. There are still some problems in the mechanism of RHA on cement hydration. For instance, the humidity gradient formed in the process of hydration makes RHA release water. But in fact, the water adsorbed by RHA is not released at one time, and the release of water should be periodic. However, the period and mechanism of water release by RHA particles are still unclear, and the research on hydration mechanism of cement-RHA system is not comprehensive and in-depth. Moreover, the research on the impact of RHA on hydration heat of cement is relatively scarce. All above need to be further studied.

3. Excessive content of K_2O in RHA may induce alkali aggregate reaction, and a small amount of K_2O increasing the alkalinity of pore solution can promote the dissolution of RHA micro-particles. However, the range in which the content of K_2O should be controlled to satisfy the requirements and its effect on the pH of pore solution and the pozzolanic reaction of SiO_2 are still in a blank stage. The research on macro-properties of cement-RHA system, such as carbonation, is still few. The comprehensive application evaluation of RHA in cement concrete needs to be studied more widely and comprehensively.

4. The contents of amorphous phase and carbon determine the quality of RHA and directly affect the pozzolanic activity. The content of amorphous phase determines the chemical reaction degree of RHA. The carbon content of RHA with insufficient combustion is higher, which has a detrimental impact on strength and several properties of cement-based materials. At present, it is still not clear which quality of RHA is more suitable for engineering applications,

and there is a lack of classification standards (such as fly ash) based on RHA quality. If RHA becomes a mineral admixture beneficial to the performance development of cement-based materials, it needs to be supported by more comprehensive application standard (such as calcination process of power plant, ash collection technology and infrastructure and technical means of RHA application, etc.) to increase the added value. That is, the higher combustion value of RHA can be fully utilized while preparing RHA. It is also necessary to comprehensively consider the environmental protection and economic benefits of RHA to ensure the green and sustainable development of RHA used in cement-based materials.

5. Carbon emission and energy consumption are the significant issues in the application of RHA in engineering. As is well known, the fresh steel slags produced by the steel mill need to be aged for a long time before it can be used as admixture of concrete. If the CO₂ from RHA production can be fully utilized to age the fresh steel slags, it can not only control the carbon emission of RHA production, but also accelerate the engineering application of the steel slags. Absolutely, this requires centralized production and a systematic production chain. The RHA has light weight, so it is easy to produce a large amount of ash in engineering application and cause secondary pollution, and the single-produced of RHA will also increase the energy consumption of production. In the previous supplementary cementitious materials, it is found that the lower limiting value of the active calcination temperature range of metakaolin coincides with the upper limit of the burning temperature of RHA. The two can be mixed and co-fired to obtain highly active composite mineral admixtures. One side, the energy consumption caused by single calcination of RHA can be avoided. On the other hand, RHA particles adhere to the metakaolin particles may avoid raising ash.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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